

CTS DEEPSKILLING ASSIGNMENTS (OUTPUT) – WEEK 1

Data structures and Algorithms

Exercise 2: E-commerce Platform Search Function

The screenshot shows the Visual Studio Code editor with the file `EcommerceSearch.java` open. The code defines a `Product` class and an `EcommerceSearch` class. The `Product` class has attributes `productId`, `productName`, and `category`. The `EcommerceSearch` class contains a `binarySearch` method that takes an array of `Product` objects and a `targetId`. The method uses a while loop to find the product with the matching ID. The terminal shows the execution of the program, where the user enters the product ID 102, and the program outputs the product details: Product Found, ID: 102, Name: Mobile, Category: Electronics.

```
1 import java.util.Scanner;
2
3 class Product {
4     int productId;
5     String productName;
6     String category;
7
8     Product(int productId, String productName, String category) {
9         this.productId = productId;
10        this.productName = productName;
11        this.category = category;
12    }
13 }
14
15 public class EcommerceSearch {
16
17     public static Product binarySearch(Product[] products, int targetId) {
18
19         int low = 0;
20         int high = products.length - 1;
21
22         while (low <= high) {
23
24             int mid = (low + high) / 2;
25
26             if (products[mid].productId == targetId)
27                 return products[mid];
28         }
29     }
30 }
```

Terminal Output:

```
PS C:\Users\Kaviyarasi\Desktop\cts_assignment> java EcommerceSearch
Enter Product ID to Search: 102
Product Found
ID: 102
Name: Mobile
Category: Electronics
PS C:\Users\Kaviyarasi\Desktop\cts_assignment> java FinancialForecasting.java
```

Exercise 7: Financial Forecasting

The screenshot shows the Visual Studio Code editor with the file `FinancialForecasting.java` open. The code defines a `FinancialForecasting` class with a `forecast` method that calculates the future value of an investment based on the initial amount, annual growth rate, and number of years. The `main` method uses a `Scanner` to take user input for these values and prints the predicted value after 5 years. The terminal shows the execution of the program, where the user enters an initial amount of 1000, an annual growth rate of 2%, and 5 years, resulting in a predicted value of 1104.08.

```
1 import java.util.Scanner;
2
3 public class FinancialForecasting {
4
5     public static double forecast(double amount, double rate, int years) {
6
7         if (years == 0) {
8             return amount;
9         }
10
11         return forecast(amount * (1 + rate), rate, years - 1);
12     }
13
14     public static void main(String[] args) {
15
16         Scanner sc = new Scanner(System.in);
17
18         System.out.print("Enter Initial Amount: ");
19         double amount = sc.nextDouble();
20
21         System.out.print("Enter Annual Growth Rate (%): ");
22         double rate = sc.nextDouble() / 100;
23
24         System.out.print("Enter Number of Years: ");
25         int years = sc.nextInt();
26     }
27 }
```

Terminal Output:

```
PS C:\Users\Kaviyarasi\Desktop\cts_assignment> java EcommerceSearch
PS C:\Users\Kaviyarasi\Desktop\cts_assignment> java FinancialForecasting.java
Enter Initial Amount: 1000
Enter Annual Growth Rate (%): 2
Enter Number of Years: 5
Predicted Value after 5 years = 1104.08
PS C:\Users\Kaviyarasi\Desktop\cts_assignment>
```